

Combining analytical techniques (HL)

Higher level (HL)

Learning outcomes

At the end of this section you should be able to interpret a variety of data, including analytical spectra, to determine the structure of a molecule.

Having pure compounds is vitally important in the manufacture of medicine and drugs. Therefore, analytical techniques are required that can assess the degree of purity in samples. Such analysis is used in the manufacture of aspirin, a medicine that has been used as a painkiller in one form or another for thousands of years (**Figure 1**). The purity of aspirin is tested using IR spectroscopy: the bonds present in a pure sample of aspirin would be 1700 cm^{-1} for the C=O bond, and around 2800 cm^{-1} for the O–H bond in carboxylic acids. The presence of these stretches and no additional stretches due to any impurities would confirm that the sample is pure.

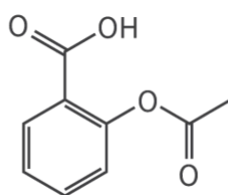


Figure 1. What stretches would be observed in the IR spectrum of aspirin? How can this be used to determine the purity of the compound?

 More information for figure 1

The image displays a chemical structure of an aspirin molecule. It shows a benzene ring attached to a carboxylic acid group on one side, represented by "C=O" and "O-H" bonds, and an ester group on the other side, represented by the "O=C-O" bonds. Aspirin's chemical structure includes these characteristic functional groups, and their presence is confirmed via infrared spectroscopy to assess purity.

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As well as measuring the purity of a compound, techniques such as IR spectroscopy can be used to identify an unknown organic compound. When identifying an organic compound, it is common to use more than one analytical technique, putting the 'clues' that each technique gives you together.

In this section, you will use each of the analytical techniques studied so far to determine the identity of an unknown organic compound. Before you start, click through to review each of the techniques covered in **Interactive 1**.

**Mass
spectrometry**

**IR
spectroscopy**

**^1H NMR
spectroscopy**

Can be used to determine
the number of different
hydrogen environments in
a compound

Can be used to determine
the relative molecular
mass of a compound

Can be used to determine
the type of bonds in a
compound

✓ Check

Interactive 1. Match the Description to the Technique.

In this example, you will determine the structure of an unknown organic compound using three different analytical techniques. The empirical formula of the compound is $\text{C}_3\text{H}_6\text{O}_2$. The mass spectrum of the unknown compound with the empirical formula of **Figure 2**. From the mass spectrum, try to determine the relative molecular mass, M_r , and the molecular formula of the compound, and then check the solution below when you are ready.

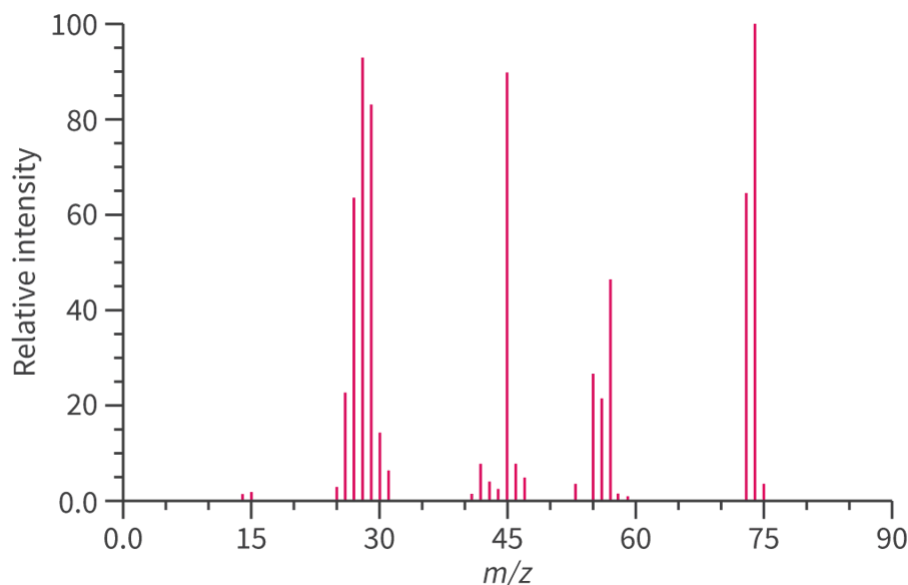


Figure 2. Use the mass spectrum to determine the relative molecular mass, M_r , of the compound.

 More information for figure 2

The image is a mass spectrum graph plotting the relative intensity against the mass-to-charge ratio (m/z). The X-axis represents m/z values ranging from 0 to 90, while the Y-axis shows relative intensity as a percentage, ranging from 0% to 100%. The graph features multiple peaks. Notable peaks include those at approximately m/z 27 with a high intensity near 100%, as well as peaks around m/z 43 and m/z 73. These peaks suggest fragment ions that can be observed at specific intervals, providing insights into the structure of the compound being analyzed. The distribution and intensity of these peaks help to determine the compound's relative molecular mass (M_r).

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The molecular ion peak occurs at m/z 74 (this is also the base ion peak). Therefore, the relative molecular mass of the compound is 74. To find the molecular formula, divide the relative molecular mass by the mass of the empirical formula and then multiply the empirical formula by that number. So for this compound we have $\frac{74}{74} = 1$, so the empirical formula and the molecular formula are the same.

Next try to determine the mass from the molecular ion that produces the fragments with m/z of 29 and m/z of 45.

m/z 45: $74 - 45 = 29$ (mass lost) this could be the loss of a $\bullet\text{CH}_2\text{CH}_3$ group

m/z 29: $74 - 29 = 45$ (mass lost) this could be the loss of a $\bullet\text{COOH}$ group

The IR spectra of the unknown compound is shown in **Figure 3**. Try to determine the bonds that produce the stretches at 1700 cm^{-1} and between 2500 and 3000 cm^{-1} .

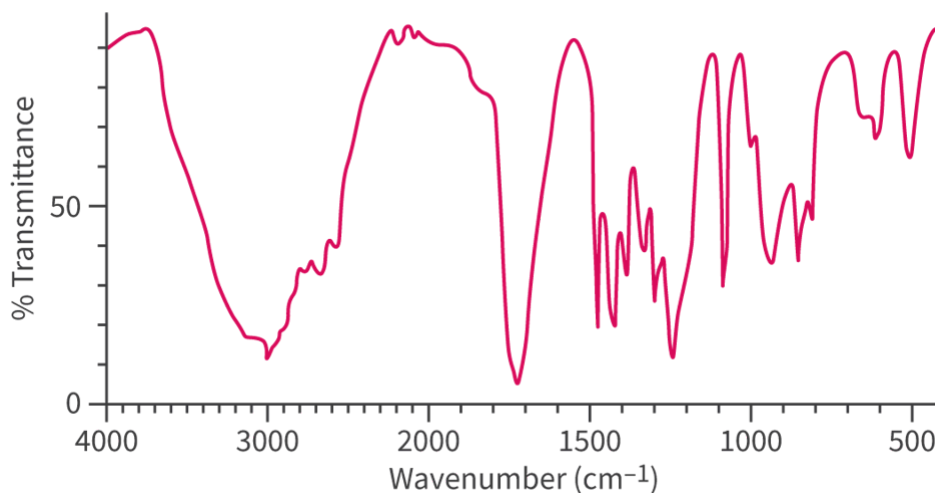


Figure 3. Use the IR spectrum to deduce the bonds that produce the stretches at 1700 cm^{-1} and between 2500 and 3000 cm^{-1} .

 More information for figure 3

The image is a graph depicting an infrared (IR) spectrum. The x-axis is labeled "Wavenumber (cm^{-1})", ranging from 4000 to 500 cm^{-1} , while the y-axis is labeled "% Transmittance", ranging from 0 to 100. The graph features a line that traverses through various peaks and troughs, indicating the transmittance levels at different wavenumbers. Notably, there is a significant trough around 1700 cm^{-1} , and a series of peaks and troughs between 2500 and 3000 cm^{-1} . Patterns show varying transmittance levels corresponding to these wavenumbers, which are critical for analyzing chemical bonds present in the compound.

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The stretch at 1700 cm^{-1} corresponds to a carbonyl group $\text{C}=\text{O}$. The stretch at $2500\text{--}3000\text{ cm}^{-1}$ corresponds to an $\text{O}\text{--}\text{H}$ bond in a carboxylic acid. From this, we can determine that the unknown compound contains a carboxyl group (COOH).

The ^1H NMR spectrum of the unknown compound is shown in **Figure 4**. Try to determine the following:

1. The number of different chemical environments that contain protons.
2. The type of proton responsible for the peak at 11.8 ppm.
3. The splitting patterns found at 1.2, 2.4 and 11.8 ppm.

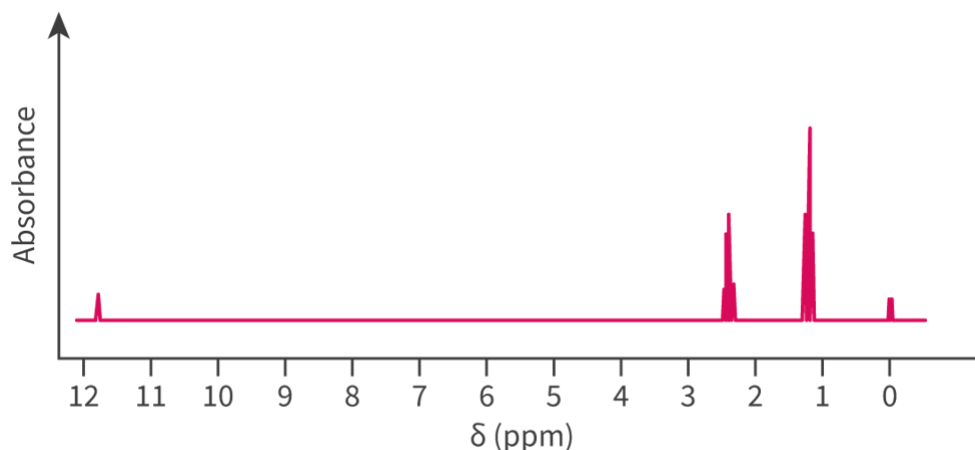


Figure 4. Use the ^1H NMR to deduce the number of different chemical environments that contain protons.

 More information for figure 4

The image is a ^1H NMR spectrum graph. The X-axis is labeled as delta (δ) in parts per million (ppm) with a range from 12 to 0 ppm, and the Y-axis is labeled as absorbance without specific units. The spectrum exhibits several peaks, with notable ones at approximately 1.0, 2.0, and 11.0 ppm. Smaller peaks are visible near 3.0 and 0.0 ppm. These peaks represent different chemical environments within a compound containing protons.

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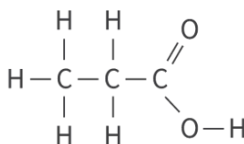
1. There are three clusters of peaks, which means there are three different chemical environments that contain protons.
2. The type of proton responsible for this peak is the H in COOH .
3. The splitting patterns are a triplet, a quartet and a singlet.

Now we will summarise the information that each technique has provided and try to deduce the structure of the compound. From the mass spectrum, we determined the relative molecular mass of the molecule, which is approximately 74, and the molecular formula, which is $\text{C}_3\text{H}_6\text{O}_2$ (the same as the empirical formula). We also deduced that the neutral fragments lost could possibly be a $\cdot\text{CH}_2\text{CH}_3$ and a $\cdot\text{COOH}$

group. From the IR spectrum, we deduced that the compound contains an OH bond in carboxylic acids and a C=O bond. From the ^1H NMR spectrum, we could deduce that there are three different chemical environments that contain protons, and the molecule has a COOH group.

Putting this all together, can you deduce the full structural formula of the compound and its IUPAC name?

The name of the compound is propanoic acid. The relative molecular mass of propanoic acid is 74.09, which is the same as the mass of the molecular ion. Propanoic acid has three different chemical environments that contain protons. The ratio of the protons is 1:2:3 and the splitting patterns are a triplet, a quartet and a singlet.



Worked example 1

In this example, you will interpret a mass spectrum, an IR spectrum and a ^1H NMR spectrum to determine the identity of an organic compound with an empirical formula of $\text{C}_3\text{H}_6\text{O}$.

1. Using the mass spectrum below, determine the following:
 - (a) The relative molecular mass and molecular formula of the compound.
 - (b) The mass lost from the molecular ion that produces the fragment at m/z 43 and the formula of the cation fragment.

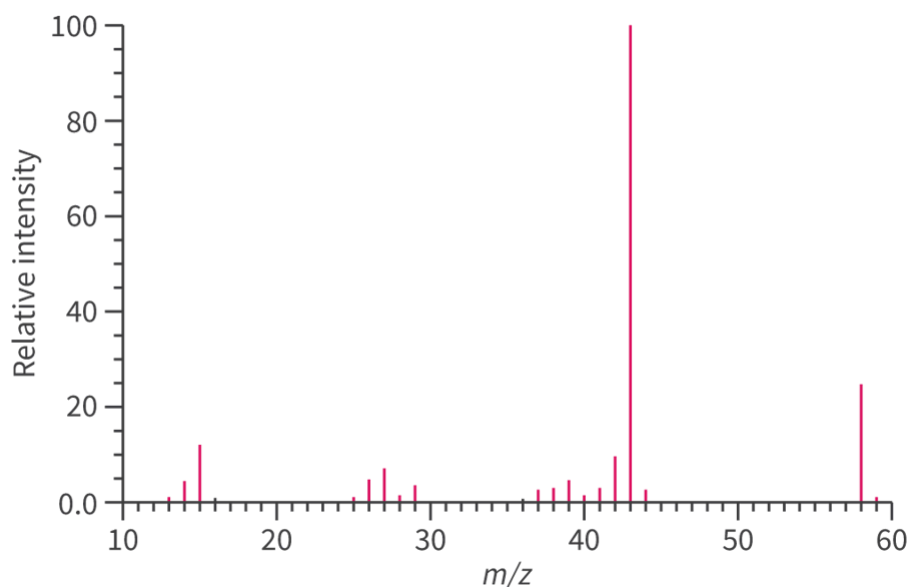


Figure 5. A mass spectrum.

 More information for figure 5

This image is a mass spectrum graph, which plots the relative intensity of ions against their mass-to-charge ratio (m/z). The X-axis represents the mass-to-charge ratio ranging approximately from 10 to 60. The Y-axis represents the relative intensity, ranging from 0 to 100. Several peaks are visible: smaller peaks can be seen at lower m/z values, while the most prominent peak is located near m/z 44, indicating the most abundant ion in the sample. Other noticeable peaks appear at m/z values around 20, 28, 32, 40, and 60, with varying intensities. The illustration is typical of mass spectrometry data used to identify compounds based on their ionized fragments.

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1. Determine the bonds that produce the stretches at 1750 cm^{-1} .

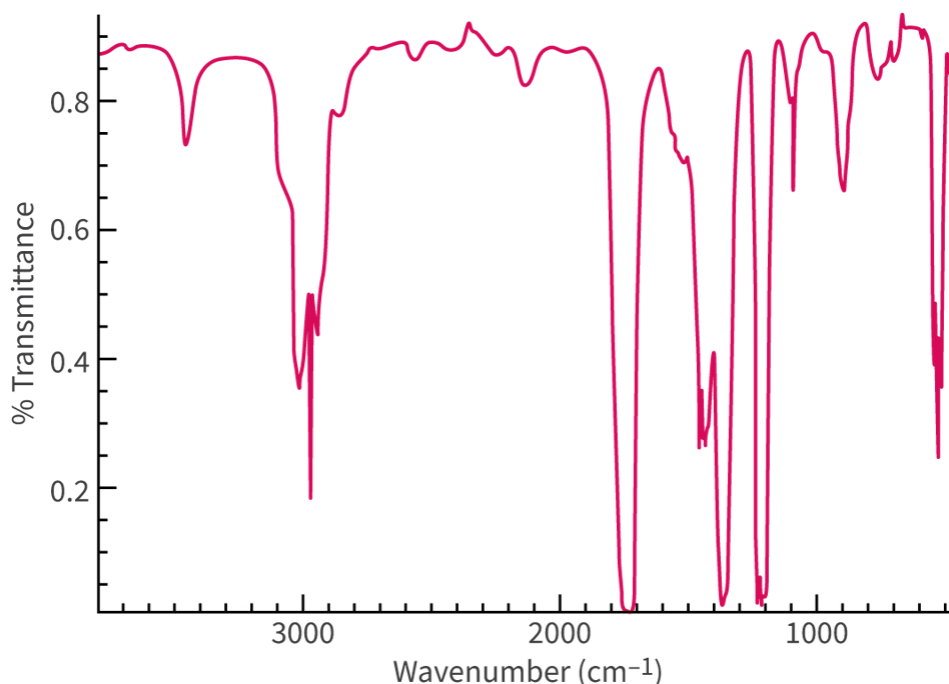


Figure 6. An IR spectrum.

 More information for figure 6

The image is a graph displaying an IR spectrum with the Y-axis labeled as '% Transmittance' and the X-axis labeled as 'Wavenumber (cm⁻¹)'. The Y-axis ranges from 0 to 1 (0% to 100% transmittance), indicating how much light passes through a sample at different energies corresponding to molecular vibrations. The X-axis ranges from approximately 4000 to 400 cm⁻¹, which is typical for IR spectroscopy, where different peaks and troughs represent absorptions at specific frequencies related to different chemical bonds.

The spectrum shows several distinct absorption peaks, including significant drops in transmittance at around 3000 cm⁻¹ and 1700 cm⁻¹, indicating strong absorption regions which are characteristic of specific molecular bonds or functional groups within the sample.

Overall, the spectrum has multiple peaks spreading from left (higher wavenumbers) to right (lower wavenumbers), demonstrating various molecular vibrations detectable in the IR region.

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1. Determine the number of different chemical environments that contain protons.
2. Deduce the full structural formula of the compound and its IUPAC name.

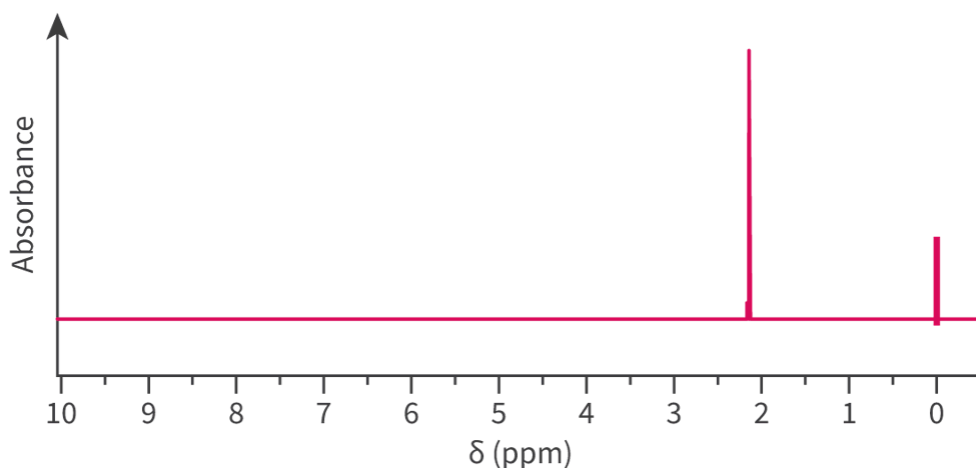


Figure 7. A ^1H NMR spectrum.

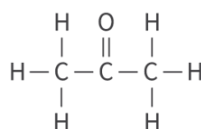
 More information for figure 7

The image displays a ^1H NMR spectrum. The horizontal X-axis represents the chemical shift in delta (δ) ppm, ranging from 0 to 10 ppm, with markings at intervals of 1 ppm. The vertical Y-axis represents absorbance, but no specific scale is provided. There are prominent peaks visible at approximately 0.9 and 2.5 ppm, indicating the presence of specific chemical environments in the sample. The baseline is flat except for these peaks.

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- (a) The m/z of the molecular ion is 58, so the relative molecular mass of the compound is approximately 58. The molecular formula is the same as the empirical formula ($\text{C}_3\text{H}_6\text{O}$).

(b) Mass lost = 15 could be a $\cdot\text{CH}_3$
Cation fragment CH_3CO^+
- The bond is a $\text{C}=\text{O}$ bond.
- There is one chemical environment that contains protons.
- The compound is propanone.



In this activity, you will practise identifying compounds using a combination of these techniques.

Activity

- **IB learner profile attribute:** Inquirers
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Try to identify a mystery compound with the empirical formula $\text{C}_3\text{H}_8\text{O}$ from the data in **Figures 8–10**.

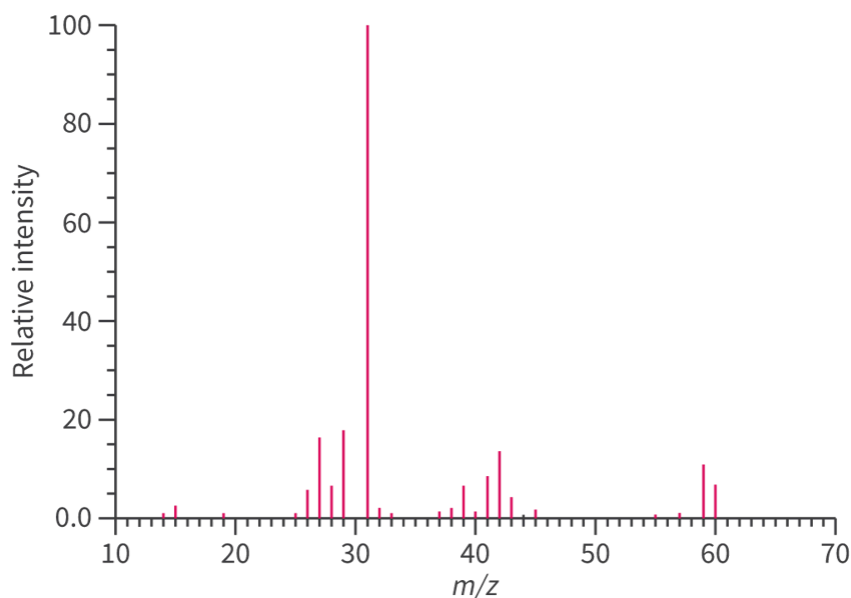


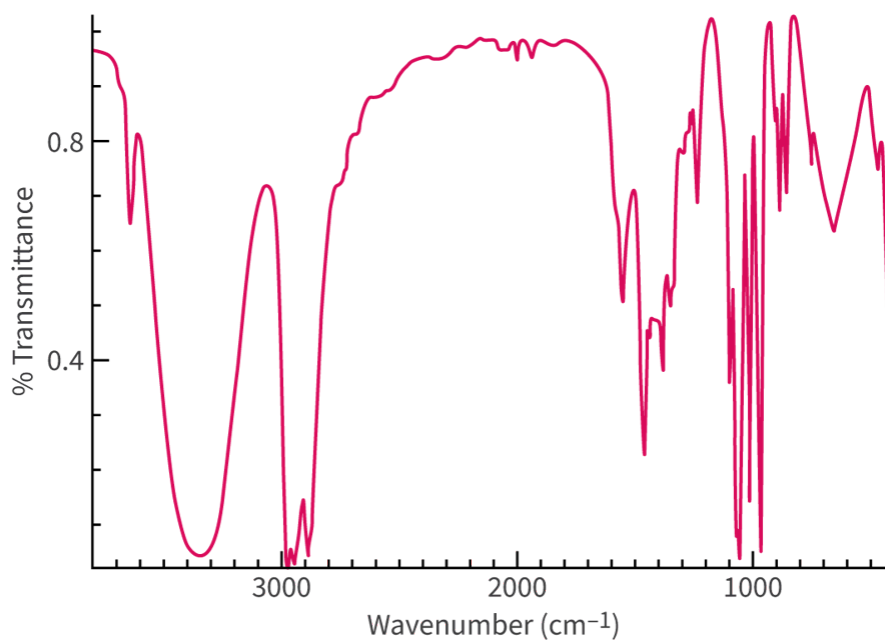
Figure 8. Mass spectrum.

 More information for figure 8

The image is a mass spectrometry graph displaying a mass spectrum. The X-axis is labeled "m/z" representing the mass-to-charge ratio, and it ranges from 10 to 70. The Y-axis is labeled "Relative intensity," ranging from 0 to 100.

Several vertical lines (peaks) are present at different m/z values, indicating the presence of ions at these ratios. The most prominent peak is at m/z 31, reaching the top of the relative intensity scale (100), suggesting it is the base peak. Other smaller peaks are visible around m/z values such as 17, 29, 43, and 59. The intensity of these peaks vary, with some being significantly shorter than others.

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**Figure 9.** IR spectrum. More information for figure 9

The image shows an IR spectrum graph with the x-axis labeled as "Wavenumber (cm^{-1})" and the y-axis labeled as "% Transmittance." The x-axis ranges from approximately 4000 cm^{-1} to 500 cm^{-1} , while the y-axis varies from 0 to 1. The spectrum exhibits multiple peaks and troughs, indicating variations in transmittance at different wavenumbers. Notably, there are significant absorptions between 3000 cm^{-1} and 1000 cm^{-1} , with a pattern of sharp declines and gradual recoveries, characteristic of IR spectra revealing the functional groups present in a sample.

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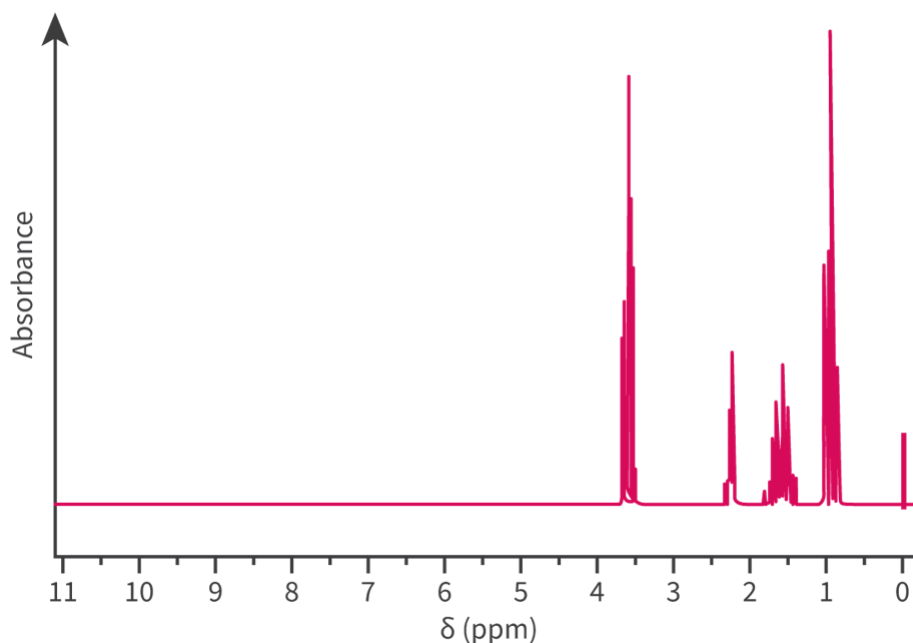


Figure 10. ^1H NMR spectrum.

 More information for figure 10.1

The image is a ^1H NMR spectrum, which is a type of graph. The X-axis represents chemical shift in parts per million (ppm), with values decreasing from 11 to 0. The Y-axis represents absorbance, but no specific scale or units are provided. Several sharp peaks of varying heights are visible in the spectrum, prominently between 0 and 5 ppm, indicating the presence of different proton environments in the sample. These peaks help in interpreting the molecular structure of the compound being analyzed.

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From the mass spectrum, we can deduce the relative molecular mass, which is 60.

From the IR spectrum, the stretch at around $3300\text{--}3400\text{ cm}^{-1}$ indicates an O—H bond in alcohols and phenols.

From the ^1H NMR spectrum, we can deduce there are four different chemical environments that contain protons.

Putting this all together, we come up with the IUPAC name of the compound, which is propan-1-ol.